Attention Is Not More Diffuse When LLMs Hallucinate: A Controlled Test of Attention Entropy as a Hallucination Detector

NeuriCo

Abstract

Attention maps come for free in a single forward pass, which makes them an attractive signal for detecting hallucination. A common intuition is that a model lacking the relevant fact attends more diffusely, so its attention entropy rises. We test this intuition directly with three pre-specified claims: late-layer attention entropy is 15–30% higher for hallucinated answers, a logistic classifier on layer-wise entropy alone reaches AUROC above 0.75, and detection is best for biography and worst for science. We run QWEN2.5-7B and LLAMA-3.1-8B on 7,500 closed-book questions from five knowledge domains, with length and domain controls and a token-likelihood baseline. The first claim is wrong in sign: final-three-layer entropy is 18.4% and 14.7% *lower* for hallucinated answers when pooled, and only 2.1% and 6.4% lower after we adjust for domain and length. The second claim holds only in a confounded setting: the entropy classifier reaches a pooled AUROC of 0.827 and 0.817, but a domain indicator alone reaches 0.785 and 0.723, and classifiers trained within a domain average 0.753 and 0.765. The third claim is half right: science is the hardest domain in both models, but biography ranks third of five. A token-likelihood baseline from the same forward pass reaches 0.876 and 0.878, and adding entropy to it gains about 0.01 AUROC. Layer-wise attention entropy carries a real but weak hallucination signal, and mixed-domain evaluations of such detectors should report a domain-only baseline and within-domain results.

1 Introduction

Large language models state wrong facts in the same fluent register as right ones. A detector that flags these errors is most useful when it is cheap enough to run on every response, which rules out methods that need several samples or an external knowledge source. Attention maps are a natural candidate: the model computes them anyway, so any signal they carry costs nothing beyond a single forward pass.

A simple and appealing intuition says what that signal should look like. When a model has the relevant fact, its attention locks onto the tokens that matter; when it does not, it "looks around" and its attention spreads out. If this is right, the Shannon entropy of attention rows should be higher when the model hallucinates, and a classifier that sees only entropy should detect hallucination.

Is attention actually more diffuse when a model hallucinates? Existing attention-based detectors do not answer this question, because they use other statistics. Lookback Lens uses the ratio of attention on context versus generated tokens [Chuang et al., 2024], LLM-Check uses attention-kernel scores [Sriramanan et al., 2024], and Yuksekgonul et al. [2024] use attention mass on constraint tokens. Other single-pass detectors probe hidden states [Azaria and Mitchell, 2023, Chen et al., 2024]

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

or use output-distribution uncertainty [Kuhn et al., 2023, Manakul et al., 2023]. What is missing is a direct, controlled test of raw attention entropy: an effect size by layer, a classifier restricted to entropy, a control for sequence length (entropy is bounded by the log of the number of attended positions), and a breakdown by domain.

We provide this test. We fix three quantitative claims before running any model: **H1**, mean attention entropy over the final three layers is 15–30% higher for hallucinated than for factual answers; **H2**, a logistic classifier on layer-wise entropy alone reaches a cross-validated AUROC above 0.75; **H3**, within-domain AUROC is highest for biography and lowest for science. We evaluate them on closed-book short-answer QA with two open-weight models, QWEN2.5-7B and LLAMA-3.1-8B, on 7,500 questions drawn evenly from five knowledge domains. We compare entropy against length-only, domain-only, and token-likelihood baselines computed from the same forward pass.

The answers are mostly negative. **H1** is wrong in sign: pooled over domains, final-three-layer entropy is 18.4% (QWEN2.5-7B) and 14.7% (LLAMA-3.1-8B) lower for hallucinated answers, and after adjusting for domain and length the difference shrinks to -2.1% and -6.4% (figure 1). **H2** passes on paper, with pooled AUROC of 0.827 and 0.817, but knowing only the domain of a question already gives 0.785 and 0.723; classifiers trained inside a single domain average 0.753 and 0.765 and range from 0.62 to 0.89. **H3** is half right: science is the hardest domain in both models, but biography is third of five. A token-likelihood baseline beats layer-wise entropy by 0.049 and 0.062 AUROC, and adding entropy to it gains only 0.015 and 0.012.

Our contributions are as follows:

- We test three pre-specified claims about attention entropy and hallucination on two model families and five knowledge domains, and report every deviation from the plan.
- We show that late-layer attention entropy is slightly lower, not higher, for hallucinated short answers, and that the sign of the difference flips between early-to-middle and upper layers.
- We show that pooled AUROC overstates the detector: a domain indicator alone recovers most of the gap between chance and the pooled entropy classifier, so we recommend a domain-only baseline and within-domain results for mixed-domain evaluations.
- We compare entropy against a token-likelihood baseline and per-head features, and find that averaging entropy over heads discards most of the usable attention signal.

We review related work in section 2, describe the setup in section 3, report results in section 4, and discuss interpretation and limitations in section 5.

2 Related Work

Output-distribution uncertainty. A first family of detectors reads uncertainty from the output distribution. Token likelihood and predictive entropy need one forward pass, while semantic entropy [Kuhn et al., 2023] and SelfCheckGPT [Manakul et al., 2023] compare several sampled responses. These methods use logits or samples, not attention. We use token-likelihood statistics from the same forward pass as our main baseline, because they are the cheapest alternative to an attention-based signal.

Hidden-state probes. A second family trains probes on internal activations. SAPLMA trains a classifier on hidden states to predict truthfulness [Azaria and Mitchell, 2023], INSIDE scores the consistency of internal representations across samples [Chen et al., 2024], and Orgad et al. [2025] study where correctness information sits in the residual stream. Unlike these methods, we restrict the detector to a scalar summary of each attention row, which tests a specific and interpretable claim rather than whatever a probe can extract.

Attention-based detectors. The closest work uses attention maps directly, but with statistics other than entropy. Lookback Lens classifies contextual hallucinations in retrieval-augmented settings from the per-head ratio of attention on context versus generated tokens [Chuang et al., 2024]. LLM-Check scores attention kernels [Sriramanan et al., 2024]. Yuksekgonul et al. [2024] relate factual errors to the attention mass placed on constraint tokens. Building on this line, we isolate the simplest version of the idea, that attention is more diffuse when the model hallucinates, and test it with explicit length and domain controls. Our finding that per-head features do much better than per-layer averages fits the head-level design of Lookback Lens.

Table 1: Data. Each domain has 1,500 questions (seed 42). The hallucination rate is the share of non-abstaining answers labeled wrong; it ranges from 23% to 94% across domains, which is the source of the domain confound in section 4.2.

		Hallucination rate	
Domain	**Source**	QWEN2.5-7B	LLAMA-3.1-8B
biography	POPQA: occupation, place of birth, father, mother	0.77	0.64
geography	POPQA: country, capital, capital of	0.38	0.33
creative works	POPQA: author, director, producer, screenwriter, composer	0.94	0.80
science	SCIQ (train split, question only)	0.23	0.27
trivia	TRIVIAQA (rc.nocontext validation)	0.42	0.25

Data and evaluation. We draw questions from POPQA [Mallen et al., 2023], SCIQ [Welbl et al., 2017], and TRIVIAQA [Joshi et al., 2017]. Because these sources differ in question style and in how often models answer correctly, we treat the source of a question as a confound to measure rather than ignore.

We did not run a systematic literature search for this study; this section covers the work we are aware of and is not exhaustive.

3 Methodology

3.1 Problem formulation

We study closed-book short-answer QA. A model receives a question q and generates an answer a with no supporting passage. We label the answer $y = 1$ (*hallucinated*) if it is factually wrong and $y = 0$ (*factual*) if it is right, and we drop abstentions. Our question is whether the attention entropy the model shows while producing a predicts y.

Hypotheses. We fixed three claims before running any model:

- **H1**: mean attention entropy over the final three layers is 15–30% higher for hallucinated than for factual answers. We count it as supported only if the relative difference and its 95% CI overlap that band with the right sign.
- **H2**: logistic regression on layer-wise entropy alone reaches a cross-validated AUROC above 0.75, pooled over domains. We count it as supported if the lower bound of the 95% CI exceeds 0.75.
- **H3**: within-domain AUROC is highest for biography and lowest for science.

3.2 Models and generation

Attention maps are required, so we use two local open-weight models: `Qwen2.5-7B-Instruct` (28 layers $\times$ 28 heads; QWEN2.5-7B) and `Llama-3.1-8B-Instruct` (32 layers $\times$ 32 heads; LLAMA-3.1-8B). We run both in float32 with eager attention. We generate with greedy decoding and at most 24 new tokens, using each model's chat template and a fixed system prompt that asks for a short answer with no explanation and a best guess (full text in section A).

3.3 Data and labels

We sample 1,500 questions per domain, 7,500 in total, with a fixed seed (table 1). Biography, geography, and creative works come from POPQA relations [Mallen et al., 2023], science from SCIQ [Welbl et al., 2017], and trivia from TRIVIAQA [Joshi et al., 2017].

Labels. We label an answer factual if a gold alias string-matches the normalized answer or a reference-based LLM judge grades it correct; otherwise we label it hallucinated. The judge is QWEN3-8B with thinking disabled and greedy decoding; it sees the question, the gold answers, and the model answer. We drop abstentions, which we identify with a regular-expression rule combined with the judge's verdict. This removes 30 examples for QWEN2.5-7B and 50 for LLAMA-3.1-8B, leaving 7,470 and 7,450.

The judge and string match agree on 92.5% (QWEN2.5-7B) and 92.1% (LLAMA-3.1-8B) of examples. Agreement is lowest in science (76% and 78%), where gold answers are free-form phrases. In a manual audit of 60 randomly chosen primary labels we found 2 clear errors and 3 debatable cases with an ambiguous question or gold answer. We repeat every headline number under judge-only and string-match-only labels.

3.4 Attention-entropy features

We run one teacher-forced forward pass over the prompt followed by the generated answer. Let $\alpha_{p,j}^{(\ell,h)}$ be the attention weight that head h in layer ℓ places from position p on position $j \le p$. For every position p that produces an answer token, the attention entropy in nats is

$$H_p^{(\ell,h)} = -\sum_{j \le p} \alpha_{p,j}^{(\ell,h)} \log \alpha_{p,j}^{(\ell,h)}. \tag{1}$$

We aggregate over positions in two ways. The primary feature, *answer mean*, averages $H_p^{(\ell,h)}$ over all answer-producing positions. The *first-token* feature uses only the position that produces the first answer token; it depends on the prompt alone, so it cannot leak answer length. A layer feature is the mean over heads, which gives $L = 28$ or 32 features per example. The **H1** statistic is the mean of the last three layer features. Because $H_p \le \log n_p$ for n_p attended positions, we also compute a length-normalized variant $H_p / \log n_p$.

3.5 Baselines and controls

We compare layer-wise entropy against the following feature sets, all fed to the same classifier:

- **Domain indicator only**: a one-hot encoding of the five domains. It measures how much a detector gains just from knowing which domain a question comes from.
- **Length only**: log prompt tokens and log answer tokens.
- **Logit baseline**: six features from the same forward pass, namely the mean, maximum, and first-token values of token negative log-likelihood and of next-token predictive entropy.
- **Combinations**: entropy + length, and entropy + logit.
- **Per-head entropy**: all $L \times H$ head-level entropies (784 or 1,024 features), as an upper bound for entropy-only detectors.

3.6 Statistics

Effect sizes. For **H1** we report the relative difference of means, $(\bar{H}_{\text{halluc}} - \bar{H}_{\text{factual}})/\bar{H}_{\text{factual}}$, with a percentile bootstrap CI from 10,000 resamples, together with Cohen's d. Per layer we run Mann–Whitney U tests with Benjamini–Hochberg correction across layers [Benjamini and Hochberg, 1995]. To adjust for confounds we fit an OLS regression of final-three-layer entropy on the hallucination label with domain fixed effects and log prompt and answer length as covariates, using HC3 standard errors.

Classifiers. Every classifier standardizes its features and fits an L2-regularized logistic regression with $C = 1$. We use 5-fold stratified cross-validation repeated 5 times, with standardization fit inside each fold, and compute AUROC on out-of-fold scores with a 2,000-resample bootstrap CI. For **H3** we train and cross-validate a separate classifier inside each domain, and we also train on one domain and test on another. All intervals are 95%.

Compute. We ran all experiments on four RTX A6000 GPUs (48 GB) shared with other jobs. Feature extraction took 24 minutes for QWEN2.5-7B and 28 minutes for LLAMA-3.1-8B at batch size 16, judging took about 30 minutes per model, and each full analysis took about 2 minutes on CPU.

Table 2: **H1**: relative difference in mean final-three-layer attention entropy, hallucinated versus factual, with 95% bootstrap CIs. Negative values mean lower entropy for hallucinated answers. No estimate comes near the hypothesized $+15\%$ to $+30\%$ band, and the large pooled difference mostly disappears within domains.

Scope	QWEN2.5-7B	LLAMA-3.1-8B
Pooled (confounded by domain)	-18.4% $[-19.4, -17.3]$, $d = -0.70$	-14.7% $[-15.4, -14.0]$, $d = -0.91$
biography	-0.8% $[-3.5, +1.8]$	-3.7% $[-5.0, -2.4]$
geography	$+2.1\%$ $[-0.7, +4.9]$	-4.5% $[-5.9, -3.0]$
creative works	-6.0% $[-11.5, -0.6]$	-14.1% $[-16.3, -11.9]$
science	-2.9% $[-5.4, -0.4]$	-4.5% $[-5.6, -3.4]$
trivia	-9.1% $[-11.7, -6.7]$	-9.8% $[-11.5, -8.0]$
OLS, domain fixed effects + length	-2.1% $[-3.2, -0.9]$	-6.4% $[-7.1, -5.7]$

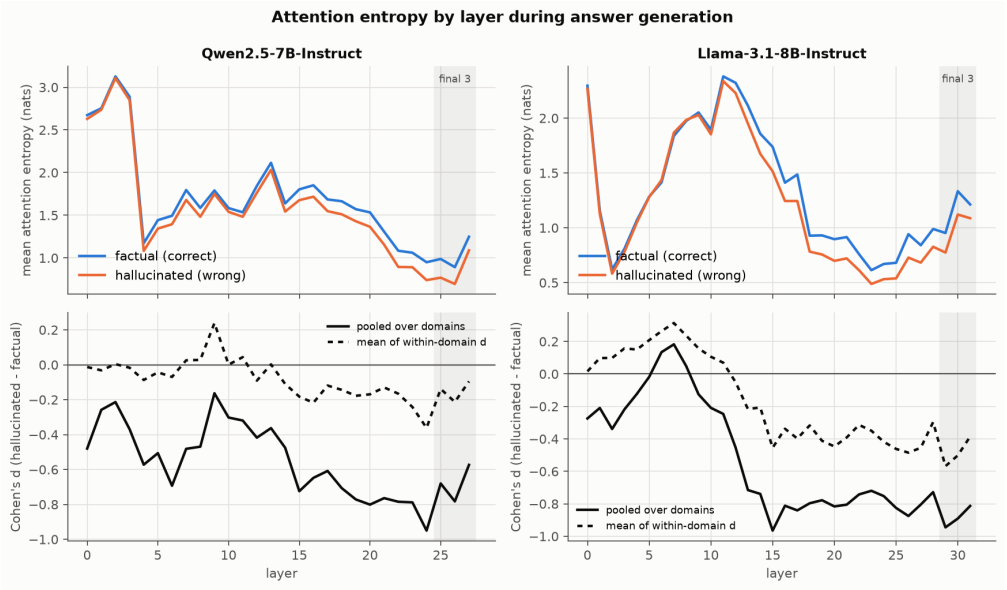


Figure 1: Attention entropy by layer during answer generation. **Top:** mean entropy for factual and hallucinated answers, pooled over domains; the shaded band marks the final three layers. **Bottom:** Cohen's d (hallucinated minus factual), pooled (solid) and averaged over within-domain estimates (dashed). Hallucinated answers have lower entropy in the upper half of both networks. Within domains, the difference is near zero or slightly positive in early-to-middle layers and much smaller than the pooled curve suggests.

4 Results

4.1 H1: entropy is lower, not higher, for hallucinated answers

Table 2 reports the relative difference in final-three-layer entropy. Pooled over domains, hallucinated answers have 18.4% lower entropy in QWEN2.5-7B (0.851 versus 1.043 nats) and 14.7% lower in LLAMA-3.1-8B (0.995 versus 1.166 nats). No estimate in either model, pooled or within a domain, comes near the hypothesized $+15\%$ to $+30\%$ band. We reject **H1**: the sign is reversed.

How much of the pooled difference is real? Most of it is a between-domain artifact. The POPQA domains have both shorter prompts with lower entropy and far more wrong answers, so pooling mixes a domain difference into the hallucination difference. Within domains, the difference ranges from -14.1% to $+2.1\%$, and two of the ten within-domain intervals include zero (figure 2). With domain fixed effects and length covariates, the adjusted difference is -2.1% $[-3.2, -0.9]$ for QWEN2.5-7B and -6.4% $[-7.1, -5.7]$ for LLAMA-3.1-8B. The effect is therefore real but small, and it is clearer in LLAMA-3.1-8B than in QWEN2.5-7B.

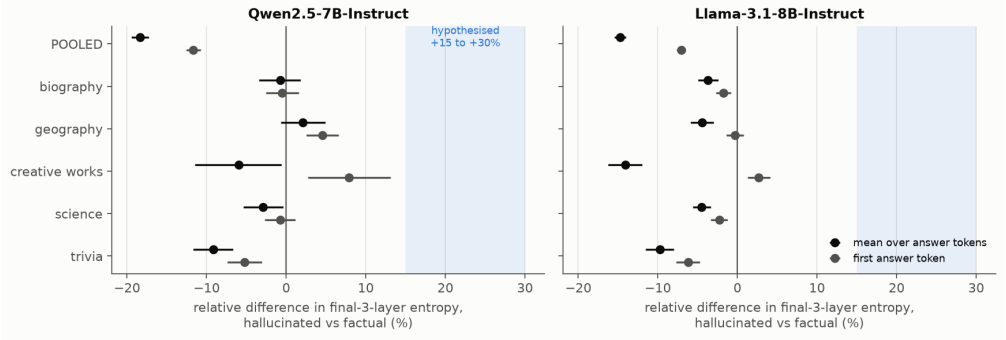


Figure 2: Relative difference in final-three-layer entropy, hallucinated versus factual, with 95% CIs, for the answer-mean feature (black) and the first-answer-token feature (gray). The shaded band is the hypothesized $+15\%$ to $+30\%$ range. Every estimate falls outside it, and the first-token feature does not even have a consistent sign across domains.

Table 3: **H2**: classifiers trained on all domains pooled. *Pooled* is the AUROC of out-of-fold scores with a 95% bootstrap CI; *Within* is the mean over the five domains of the AUROC of the same scores computed inside each domain. A domain indicator alone reaches 0.785 / 0.723 pooled, so pooled numbers overstate every detector. Best result per column in **bold**.

	QWEN2.5-7B		LLAMA-3.1-8B	
Features	Pooled	Within	Pooled	Within
Controls				
domain indicator only	0.785 [0.774, 0.796]	0.429	0.723 [0.711, 0.736]	0.423
length only	0.669 [0.657, 0.681]	0.527	0.652 [0.639, 0.664]	0.535
Attention entropy				
final-3-layer mean (1 feature)	0.700 [0.688, 0.711]	0.539	0.731 [0.720, 0.742]	0.638
layer-wise, answer mean (primary)	0.827 [0.817, 0.836]	0.719	0.817 [0.807, 0.826]	0.740
layer-wise, first answer token	0.826 [0.816, 0.835]	0.683	0.821 [0.812, 0.830]	0.743
layer-wise, length-normalized	0.826 [0.817, 0.835]	0.718	0.815 [0.806, 0.825]	0.740
layer-wise + length	0.833 [0.824, 0.842]	0.727	0.822 [0.813, 0.831]	0.747
per-head ($L \times H$ features)	0.890 [0.883, 0.898]	0.821	0.874 [0.867, 0.882]	0.830
Token likelihood				
logit baseline	0.876 [0.868, 0.884]	0.812	0.878 [0.870, 0.886]	0.847
layer-wise entropy + logit	**0.891** [0.884, 0.899]	**0.823**	**0.890** [0.883, 0.898]	**0.854**

Robustness. The result does not depend on our analysis choices. Length normalization changes nothing material: the pooled difference is -17.6% and -14.1%. Label choice does not matter either: the pooled difference is -17.9% and -14.2% with judge-only labels and -15.3% and -12.7% with string-match labels. The first-token feature is the one variant that behaves differently. Its within-domain difference ranges from -6.2% to $+7.9\%$, with a domain average of $+1.2\%$ for QWEN2.5-7B and -1.6% for LLAMA-3.1-8B, so at the moment the model starts its answer there is no consistent entropy shift in the final three layers.

Layer profile. Figure 1 shows why a single late-layer average is a poor summary. Entropy differences are near zero or slightly positive in early-to-middle layers and negative in the upper half of the network. In LLAMA-3.1-8B, the within-domain effect peaks at $d \approx +0.3$ around layer 7 and falls to between -0.3 and -0.5 in the upper layers. Pooled, the difference is significant after FDR correction in 28 of 28 layers for QWEN2.5-7B and 30 of 32 for LLAMA-3.1-8B, but this pooled significance inherits the domain confound.

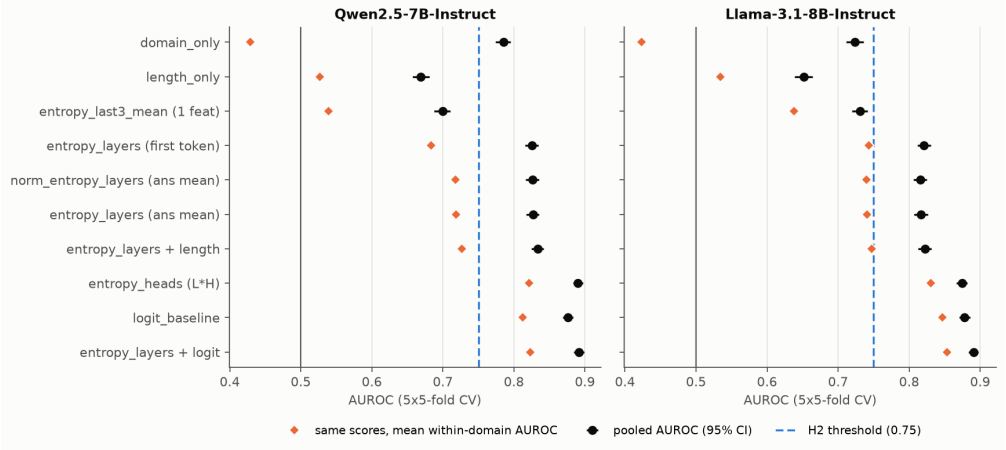


Figure 3: AUROC by feature set for classifiers trained on pooled data. Black points are pooled AUROC with 95% CIs; orange diamonds are the mean within-domain AUROC of the same scores; the dashed line is the **H2** threshold of 0.75. Every pooled number sits well above its within-domain counterpart, and the domain indicator alone clears or approaches the threshold.

4.2 H2: pooled AUROC clears 0.75, but mostly for the wrong reason

By the pre-specified criterion, **H2** passes in both models: layer-wise entropy reaches a pooled AUROC of 0.827 [0.817, 0.836] for QWEN2.5-7B and 0.817 [0.807, 0.826] for LLAMA-3.1-8B (table 3). The result is stable: the standard deviation of AUROC across the five cross-validation repeats is below 0.001.

Does this number mean what it appears to mean? Not quite. A classifier that sees only which of the five domains a question comes from already reaches 0.785 and 0.723, because hallucination rates range from 23% to 94% across domains (table 1) and layer-wise entropy encodes domain. Layer-wise entropy beats this domain-only control by $+0.041$ [0.032, 0.051] for QWEN2.5-7B and $+0.093$ [0.082, 0.105] for LLAMA-3.1-8B. When we score the same out-of-fold predictions inside each domain, the mean AUROC drops to 0.719 and 0.740 (figure 3). We therefore read **H2** as mixed: the criterion is met, but most of the margin over chance could be obtained from the domain alone.

The hypothesized feature is weak. The specific statistic in **H1** and **H2**, the mean entropy of the final three layers, reaches only 0.700 and 0.731 pooled and 0.539 and 0.638 within domain. The full layer profile is needed to get a usable detector.

Entropy is not a length artifact. Layer-wise entropy beats the length-only control by $+0.158$ [0.145, 0.170] and $+0.165$ [0.152, 0.178] AUROC. Normalizing entropy by $\log n_p$ or adding length features changes pooled AUROC by less than 0.01.

Token likelihood is a stronger signal. The logit baseline reaches 0.876 and 0.878, which is 0.049 [0.040, 0.058] and 0.062 [0.053, 0.070] above layer-wise entropy. Adding entropy to the logit baseline gains $+0.015$ [0.010, 0.020] and $+0.012$ [0.009, 0.016]. The gain is statistically reliable but small.

Averaging over heads loses signal. Per-head entropy reaches 0.890 and 0.874 pooled and 0.821 and 0.830 within domain, about 0.10 above per-layer entropy within domain and comparable to the logit baseline. Most of the usable attention signal lives in individual heads and is lost when we average a layer.

4.3 H3: science is hardest, but biography is not easiest

Table 4 reports classifiers trained and cross-validated inside each domain. Layer-wise entropy averages 0.753 for QWEN2.5-7B and 0.765 for LLAMA-3.1-8B, with a range of 0.616 to 0.892. Only geography and creative works exceed 0.75 in both models.

Science is lowest. Science has the lowest AUROC in both models, 0.616 and 0.694. The bootstrap probability that science is the lowest domain is 1.00 for QWEN2.5-7B and 0.98 for LLAMA-3.1-

Table 4: **H3**: AUROC of classifiers trained and cross-validated inside each domain. The entropy rows carry 95% bootstrap CIs. Science is the lowest domain for layer-wise entropy in both models; biography is third. Best domain per row in **bold**.

	Features	biography	geography	creative works	science	trivia	mean
QWEN2.5-7B	layer-wise entropy	0.747 [0.716, 0.775]	0.802 [0.777, 0.824]	**0.892** [0.857, 0.922]	0.616 [0.581, 0.648]	0.711 [0.684, 0.736]	0.753
	length only	0.507	0.599	0.552	0.546	0.593	0.559
	logit baseline	0.781	0.800	**0.911**	0.728	0.853	0.815
	entropy + logit	0.810	0.860	**0.932**	0.730	0.839	0.834
LLAMA-3.1-8B	layer-wise entropy	0.742 [0.713, 0.768]	**0.831** [0.808, 0.851]	0.820 [0.792, 0.846]	0.694 [0.664, 0.722]	0.741 [0.711, 0.769]	0.765
	length only	0.423	**0.655**	0.633	0.584	0.491	0.557
	logit baseline	0.827	0.853	**0.885**	0.792	0.879	0.847
	entropy + logit	0.845	0.877	**0.896**	0.813	0.883	0.863

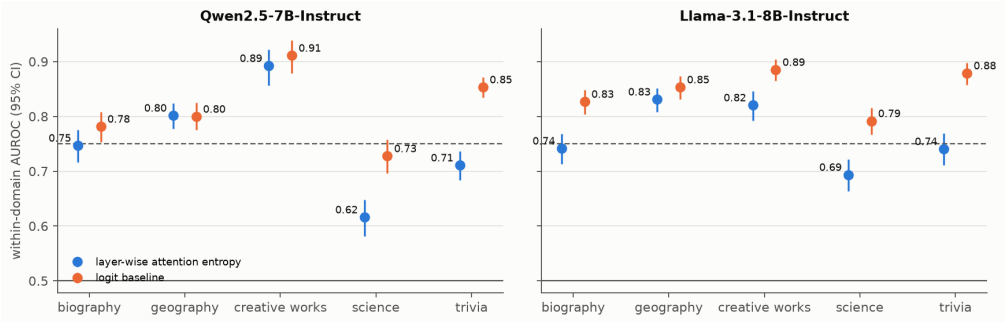


Figure 4: Within-domain AUROC with 95% CIs for layer-wise attention entropy (blue) and the logit baseline (orange). The dashed line marks 0.75. Science is the lowest domain for entropy in both models, and the logit baseline dips there too. Entropy trails the logit baseline most on trivia.

8B, and the ordering holds under all three labelings. The logit baseline shows the same dip (0.728 and 0.792), so the weakness in science is not specific to attention features (figure 4).

Combining entropy with the logit baseline raises the within-domain mean from 0.815 to 0.834 for QWEN2.5-7B and from 0.847 to 0.863 for LLAMA-3.1-8B, but the gain is not uniform: on trivia for QWEN2.5-7B the combination (0.839) is below the logit baseline alone (0.853).

Biography is not highest. Biography ranks third of five in both models under the primary labels, behind creative works and geography, and it is never the top domain in any bootstrap resample. In LLAMA-3.1-8B, biography and trivia are effectively tied (0.742 versus 0.741), and biography falls to fourth under both alternative labelings. Biography does beat science, by $+0.131$ [0.083, 0.176] in QWEN2.5-7B and $+0.048$ [0.009, 0.088] in LLAMA-3.1-8B. **H3** is thus half supported.

Cross-domain transfer. Training on one domain and testing on another gives a mean off-diagonal AUROC of 0.651 for QWEN2.5-7B and 0.682 for LLAMA-3.1-8B, against 0.753 and 0.765 within domain (figure 5). Part of the signal generalizes across domains and part is domain-specific. Transfer is uneven: in QWEN2.5-7B, classifiers trained on any other domain reach 0.73 to 0.88 on creative works, while no classifier trained elsewhere exceeds 0.60 on science.

4.4 Error analysis

By relation. Within POPQA, the pooled entropy classifier is strongest on *capital* (0.85 for QWEN2.5-7B, 0.84 for LLAMA-3.1-8B) and, for QWEN2.5-7B, on *author* and *screenwriter* (0.88 to 0.91). For QWEN2.5-7B it is weakest on *father*, *place of birth*, *country*, and *capital of* (0.66 to 0.68). Table 6 in the appendix lists all relations.

Missed hallucinations. Misses cluster in science and trivia questions where the model gives a fluent, confident near-miss, such as "Bone marrow" for the skeleton, or "pharaoh" for the Sphinx's human head. These answers often have very low token negative log-likelihood as well, so token likelihood does not flag them either.

False alarms. False alarms are mostly correct answers about obscure POPQA entities, such as the screenwriter of *A Special Day*, where the question looks like the ones the model usually gets wrong.

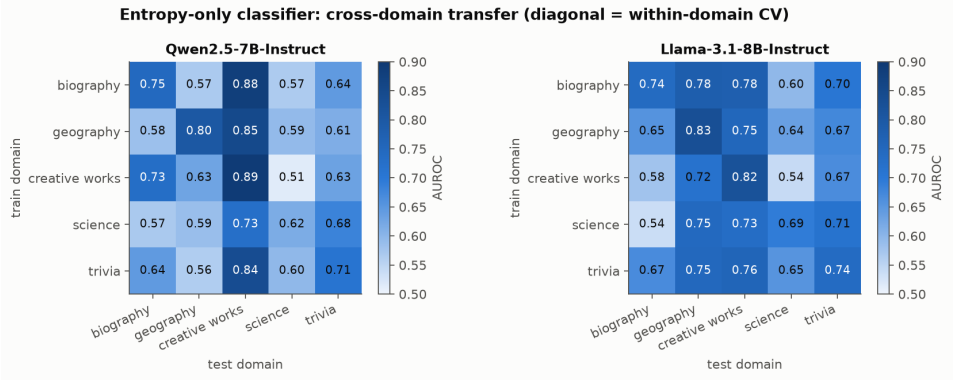


Figure 5: Cross-domain transfer of the layer-wise entropy classifier. Rows are training domains and columns are test domains; the diagonal is within-domain cross-validated AUROC. Off-diagonal cells average 0.651 (QWEN2.5-7B) and 0.682 (LLAMA-3.1-8B), so the signal transfers only partly. Science is the hardest test domain to transfer into in both models.

Label noise. Some apparent misses are label errors: "Belgium" for the origin of French fries is graded wrong against the gold answer.

Overlap with token likelihood. Entropy and logit scores are correlated but not redundant, with a within-domain Spearman correlation of 0.33 for QWEN2.5-7B and 0.50 for LLAMA-3.1-8B. This is consistent with the small gain from combining them.

5 Discussion

The direction of the effect. In the upper layers, attention is slightly *more* concentrated when the answer is wrong, the opposite of the "looking around" intuition. One possible reading is that when the model lacks the fact, upper-layer heads have less relevant material to integrate and fall back on a few fixed positions, whereas a retrieved fact spreads attention over the subject and relation tokens. We did not test this mechanism. It is a hypothesis for follow-up work, not a finding.

Why pooled numbers mislead. Domains differ at once in prompt style, in attention entropy, and in the base rate of error. Any feature that separates domains will therefore look like a hallucination detector on pooled data. In our data a five-way domain indicator reaches 0.72 to 0.79 AUROC with no information about the answer at all. We recommend that studies of this kind report a domain-only baseline and a within-domain breakdown by default.

Relation to prior work. Our results agree with the general picture that single-pass internal signals carry correctness information, but that output probabilities are a strong baseline for short-answer QA. That per-head features do much better than per-layer averages fits the head-level approach of Lookback Lens [Chuang et al., 2024].

Practical implication. Layer-wise attention entropy is not a good stand-alone hallucination detector for short factual answers. Token likelihood from the same forward pass is stronger, and entropy adds about 0.01 AUROC on top of it.

5.1 Limitations

Domain is confounded with dataset. Science and trivia come from different sources than the three POPQA domains, with different question styles and label quality. **H3** comparisons that involve science may reflect SCIQ rather than science as a subject.

Label noise. By our 60-example audit, roughly 3–8% of labels are wrong or debatable, and judge and string match disagree most in science. Noise lowers every AUROC and could contribute to the science dip. Results agree across all three labelings.

Deviations from the plan. We made several changes after the plan was written, listed in full in section B. Three matter for interpretation. First, the judge is an 8B model rather than the planned

14B model. Second, we redefined the primary label as a hybrid of judge and string match after auditing judge errors; we made this choice after seeing provisional string-match-only results but before any judge-based result. Third, we added the domain-only control and the within-domain reading of **H2** after a provisional run exposed the confound, so that part of the analysis is not pre-specified.

Task scope. We study short-answer closed-book QA with answers of a few tokens. We did not test long-form generation, summarization, or retrieval-augmented settings, and entropy during generation may behave differently there.

Models. We test two instruction-tuned models of 7–8B parameters and no larger models.

Small factual class. QWEN2.5-7B answers only 88 creative-works questions correctly, so estimates for that domain are wide.

Feature definition. We take entropy over all attended positions, including chat-template and system-prompt tokens, and did not examine entropy restricted to question tokens. We fixed classifier regularization at $C = 1$ and did not tune it.

6 Conclusion

We tested the intuition that attention is more diffuse when a language model hallucinates. On closed-book short-answer QA with two open-weight models and five knowledge domains, hallucinated answers do not show 15–30% higher late-layer attention entropy; they show slightly lower entropy, about 2–6% after we control for domain and length. A layer-wise entropy classifier reaches about 0.82 AUROC pooled, but only 0.62 to 0.89 within a domain, and it stays below a simple token-likelihood baseline. Science is the hardest domain as predicted; biography is not the easiest.

The main takeaway is methodological: on mixed-domain data, a feature that encodes the domain can pass for a hallucination detector, so evaluations should report a domain-only baseline and within-domain results.

Several follow-ups would sharpen these conclusions. Drawing all domains from one source, or matching question templates, would break the confound between domain and dataset. Long-form generation with claim-level labels would test entropy during generation where it is a richer signal. Per-head features recover about 0.10 AUROC over per-layer means, so identifying which heads carry the signal, and whether they are the same heads across domains, is a natural next step. Finally, restricting entropy to subject or question tokens and relabeling science with a stronger judge or human annotation would address two limitations of the current study.

References

Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it's lying. In *Findings of EMNLP*, 2023.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300, 1995.

Chao Chen, Kai Liu, Ze Chen, Yi Gu, Yue Wu, Mingyuan Tao, Zhihang Fu, and Jieping Ye. INSIDE: LLMs' internal states retain the power of hallucination detection. In *International Conference on Learning Representations*, 2024.

Yung-Sung Chuang, Linlu Qiu, Cheng-Yu Hsieh, Ranjay Krishna, Yoon Kim, and James Glass. Lookback lens: Detecting and mitigating contextual hallucinations in large language models using only attention maps. In *Proceedings of EMNLP*, 2024.

Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of ACL*, 2017.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations*, 2023.

Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of ACL*, 2023.

Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of EMNLP*, 2023.

Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, Hadas Kotek, and Yonatan Belinkov. LLMs know more than they show: On the intrinsic representation of LLM hallucinations. In *International Conference on Learning Representations*, 2025.

Gaurang Sriramanan, Siddhant Bharti, Vinu Sankar Sadasivan, Shoumik Saha, Priyatham Kattakinda, and Soheil Feizi. LLM-Check: Investigating detection of hallucinations in large language models. In *Advances in Neural Information Processing Systems*, 2024.

Johannes Welbl, Nelson F. Liu, and Matt Gardner. Crowdsourcing multiple choice science questions. In *Proceedings of the 3rd Workshop on Noisy User-generated Text*, 2017.

Mert Yuksekgonul, Varun Chandrasekaran, Erik Jones, Suriya Gunasekar, Ranjita Naik, Hamid Palangi, Ece Kamar, and Besmira Nushi. Attention satisfies: A constraint-satisfaction lens on factual errors of language models. In *International Conference on Learning Representations*, 2024.

A Implementation details

System prompt. Both subject models receive the following system prompt:

> You are a knowledgeable assistant. Answer the question with only the short answer (a name or a few words). Do not explain. Always give your best guess.

Models. The subject models are `Qwen/Qwen2.5-7B-Instruct` and `unsloth/Meta-Llama-3.1-8B-Instruct`, both in float32 with eager attention. The judge is `Qwen/Qwen3-8B` in bfloat16 with thinking disabled, greedy decoding, and 4 new tokens. Mean prompt length is 54 to 64 tokens for QWEN2.5-7B and 76 to 86 tokens for LLAMA-3.1-8B, depending on the domain; mean answer length is 2.5 to 4.0 and 3.4 to 4.4 tokens.

B Deviations from the plan

We wrote the analysis plan before running any model. We deviated from it in five ways.

1. **Precision.** Half precision produced NaN attention rows with left-padded batches on QWEN2.5-7B, so we ran both subject models in float32.
2. **Judge model.** The planned judge, `Qwen2.5-14B-Instruct`, did not fit in memory on the shared GPUs. A 4-bit version was too slow and we stopped it after 128 verdicts. We used QWEN3-8B instead. On the 128 overlapping items the two judges agree on 123.
3. **Primary label.** An audit of disagreements between judge and string match showed that the 8B judge errs in both directions, for example grading an answer incorrect although it appears verbatim in the gold list. We therefore changed the primary label to a hybrid: correct if a gold alias string-matches or the judge says correct, with a regular-expression abstention rule on top. We made this decision after seeing provisional string-match-only results but before any judge-based result.
4. **Added controls.** We added the domain-only classifier, entropy + domain, and the within-domain AUROC of pooled scores after a provisional run showed that domain identity alone predicts correctness well.
5. **Bootstrap size.** AUROC confidence intervals use 2,000 resamples.

C Additional results

Table 5 repeats the headline numbers under the three labelings, and table 6 breaks the pooled classifiers down by POPQA relation.

Table 5: Label robustness. *Effect* is the pooled relative difference in final-three-layer entropy; *Pooled* and *Within* are the AUROC of the layer-wise entropy classifier trained on pooled data, scored on all data and averaged within domains. Conclusions do not depend on the labeling.

	QWEN2.5-7B			LLAMA-3.1-8B		
Labels	Effect	Pooled	Within	Effect	Pooled	Within
hybrid (primary)	-18.4%	0.827	0.719	-14.7%	0.817	0.740
judge only	-17.9%	0.822	0.715	-14.2%	0.809	0.728
string match only	-15.3%	0.793	0.716	-12.7%	0.792	0.736

Table 6: AUROC by POPQA relation, using out-of-fold scores of the classifiers trained on pooled data. *Rate* is the hallucination rate. A dash marks relations with fewer than 20 examples in the smaller class, which we do not score. The logit baseline is at least as good as entropy on every relation except *author* and *screenwriter* for QWEN2.5-7B.

		QWEN2.5-7B			LLAMA-3.1-8B		
Relation	**Domain**	Rate	Entropy	Logit	Rate	Entropy	Logit
occupation	biography	0.72	0.746	0.764	0.56	0.675	0.772
place of birth	biography	0.87	0.666	0.793	0.79	0.684	0.798
father	biography	0.72	0.656	0.821	0.56	0.761	0.863
mother	biography	0.74	0.726	0.791	0.64	0.722	0.903
country	geography	0.40	0.671	0.774	0.39	0.812	0.836
capital	geography	0.24	0.851	0.893	0.16	0.843	0.912
capital of	geography	0.56	0.684	0.792	0.47	0.667	0.757
author	creative works	0.88	0.905	0.892	0.74	0.795	0.890
screenwriter	creative works	0.94	0.876	0.868	0.80	0.785	0.885
director	creative works	–	–	–	0.77	0.781	0.858
producer	creative works	–	–	–	0.90	0.765	0.896
composer	creative works	–	–	–	0.81	0.749	0.904